

Mathematical Analysis 2

Seminar 6

Improper integrals

$$\begin{aligned}\int_a^\infty f(x) dx &\stackrel{\text{def}}{=} \lim_{b \rightarrow \infty} \int_a^b f(x) dx. \\ \int_{-\infty}^b f(x) dx &\stackrel{\text{def}}{=} \lim_{a \rightarrow -\infty} \int_a^b f(x) dx \\ \int_{-\infty}^\infty f(x) dx &\stackrel{\text{def}}{=} \lim_{\substack{a \rightarrow -\infty \\ b \rightarrow \infty}} \int_a^b f(x) dx.\end{aligned}$$

 **Example 1** $> \int_0^\infty e^{-x} dx = \lim_{b \rightarrow \infty} (-e^{-x}) \Big|_0^b = \lim_{b \rightarrow \infty} (1 - e^{-b}) = 1;$

 **Example 2** $> \int_1^\infty \frac{dx}{x} = \ln x \Big|_1^\infty = \infty \quad (\text{divergent});$

 **Example 3** $> \int_0^\infty \sin x dx$
 $\int_0^b \sin x dx = 1 - \cos b$; $\cos$ has no limit at ∞ ; the integral is divergent.

$$\begin{aligned}\int_a^{b-0} f(x) dx &:= \lim_{\varepsilon \rightarrow 0+} \int_a^{b-\varepsilon} f(x) dx, \quad b \text{ is a singular point for } f; \\ \int_{a+0}^b f(x) dx &\stackrel{\text{def}}{=} \lim_{\varepsilon \rightarrow 0+} \int_{a+\varepsilon}^b f(x) dx, \quad a \text{ is a singular point for } f; \\ \int_{a+0}^{b-0} f(x) dx &\stackrel{\text{def}}{=} \lim_{\varepsilon_1, \varepsilon_2 \rightarrow 0+} \int_{a+\varepsilon_1}^{b-\varepsilon_2} f(x) dx, \quad a \text{ and } b \text{ are singular points for } f; \\ \int_a^b f(x) dx &\stackrel{\text{def}}{=} \int_a^{c-0} f(x) dx + \int_{c+0}^b f(x) dx, \quad c \in (a, b) \text{ is a singular point for } f.\end{aligned}$$

For simplicity, write

$$\int_a^b f(x) dx \text{ instead of } \int_{a+0}^b f(x) dx, \int_a^{b-0} f(x) dx, \int_{a+0}^{b-0} f(x) dx.$$

 **Example 4** > $\int_0^1 \frac{dx}{\sqrt{x}} = \lim_{\varepsilon \rightarrow 0+} \int_{\varepsilon}^1 \frac{dx}{\sqrt{x}} = \lim_{\varepsilon \rightarrow 0+} 2\sqrt{x} \Big|_{\varepsilon}^1 = \lim_{\varepsilon \rightarrow 0+} (2 - 2\sqrt{\varepsilon}) = 2$
(convergent);

 **Example 5** > $\int_0^1 \frac{dx}{x} = \ln x \Big|_{0+}^1 = 0 - \lim_{\varepsilon \rightarrow 0+} \ln \varepsilon = \infty$ (divergent);

 **Example 6** > $\int_1^{\infty} \frac{1}{x^p} dx$, with $p \in \mathbb{R}$.

$$\int_1^{\infty} \frac{1}{x^p} dx = \begin{cases} \frac{1}{p-1}, & \text{dacă } p > 1 \quad (\text{convergent}), \\ \infty, & \text{dacă } p \leq 1 \quad (\text{divergent}). \end{cases}$$

 **Example 7** > $\int_0^1 \frac{1}{x^p} dx$, with $p \in \mathbb{R}$. The singular point is 0 (if $p > 0$).

$$\int_0^1 \frac{1}{x^p} dx = \begin{cases} \frac{1}{1-p}, & p < 1 \quad (\text{convergent}), \\ \infty, & p \geq 1 \quad (\text{divergent}). \end{cases}$$

Remark. (splitting an improper integral with multiple singular points – an example)

$$\begin{aligned} \int_0^{\infty} \frac{e^{-x}}{\sqrt{x \cdot |x-2|}} dx &= \int_{0+0}^1 \frac{e^{-x}}{\sqrt{x(2-x)}} dx + \int_1^{2-0} \frac{e^{-x}}{\sqrt{x(2-x)}} dx \\ &+ \int_{2+0}^3 \frac{e^{-x}}{\sqrt{x(x-2)}} dx + \int_3^{\infty} \frac{e^{-x}}{\sqrt{x(x-2)}} dx \end{aligned}$$

Convergence criteria for improper integrals of positive functions

Theorem 1 (comparison). Let $I \subseteq \mathbb{R}$ be an interval (e.g., $[a, \infty)$, $[a, b]$, $(-\infty, b]$, $(a, b]$, (a, b) , $(-\infty, \infty)$, ...) and let $f, g : I \rightarrow \mathbb{R}_+$ locally integrable such that

$$0 \leq f(x) \leq g(x), \quad \text{for all } x \in I.$$

- (i) If $\int_I g(x) dx$ is convergent, then $\int_I f(x) dx$ is convergent.
- (ii) If $\int_I f(x) dx$ is divergent, then $\int_I g(x) dx$ is divergent.

Theorem 2 (C1). Let $f : [a, \infty) \rightarrow \mathbb{R}_+$ be locally integrable. Let $p \in \mathbb{R}$ such that

$$\lim_{x \rightarrow \infty} x^p \cdot f(x) = L \in [0, \infty].$$

- (i) If $p > 1$ and $L \neq \infty$, then $\int_a^{\infty} f(x) dx$ is convergent.
- (ii) If $p \leq 1$ and $L \neq 0$, then $\int_a^{\infty} f(x) dx$ is divergent.

Example 8 $\int_2^\infty \frac{dx}{x\sqrt{1+x^4}}$

$f(x) = \frac{1}{x\sqrt{1+x^4}}$ well defined on $[2, \infty)$
 is continuous continuous
 is ≥ 0

$L = \lim_{x \rightarrow \infty} x^p \cdot \frac{1}{x\sqrt{1+x^4}} = \lim_{x \rightarrow \infty} \frac{x^{p-1}}{\sqrt{1+x^4}} = \lim_{x \rightarrow \infty} \frac{x^2}{\sqrt{1+x^4}} \cdot \lim_{x \rightarrow \infty} x^{p-3}$

① We should choose p such that $L \in (0, \infty)$ ($L \neq 0, L \neq \infty$)
 $p=3 \Rightarrow L = \lim_{x \rightarrow \infty} \frac{x^2}{\sqrt{1+x^4}} = 1$ ($\frac{x^2}{x^2\sqrt{\frac{1}{x^4}+1}} = \frac{1}{\sqrt{1+\frac{1}{x^4}}} \rightarrow \frac{1}{\sqrt{1+0}}$)
 $p=3 > 1, L \in (0, \infty) \xrightarrow{C1} \text{convergence}$

Problem 1: $\int_0^\infty \frac{x^{\sqrt{2}}}{1+x^2} dx$

$L = \lim_{x \rightarrow \infty} x^p \cdot f(x) = \lim_{x \rightarrow \infty} \frac{x^{p+\sqrt{2}}}{x^2+1} = \lim_{x \rightarrow \infty} \frac{x^2}{x^2+1} = 1$

$p+\sqrt{2}=2$
 $p=2-\sqrt{2} \leq 1 \xrightarrow{C1} \text{divergent}$

Problem 2: $\int_1^\infty \frac{\arctan x}{x\sqrt{x}} dx$

$L = \lim_{x \rightarrow \infty} x^p \cdot \frac{\arctan x}{x\sqrt{x}} = \lim_{x \rightarrow \infty} \frac{x^p \arctan x}{x^{3/2}}$
 $p = \frac{3}{2} \Rightarrow L = \frac{\pi}{2}$; $p > 1 \xrightarrow{C1} \text{convergent}$

Theorem 3 (C2). Let $f : (a, b] \rightarrow \mathbb{R}_+$ be locally integrable. Let $p \in \mathbb{R}$ such that
 $\lim_{\substack{x \rightarrow a \\ x > a}} (x-a)^p \cdot f(x) = L \in [0, \infty]$.

(i) If $p < 1$ and $L \neq \infty$, then $\int_a^\infty f(x) dx$ is convergent.

(ii) If $p \geq 1$ and $L \neq 0$, then $\int_a^\infty f(x) dx$ is divergent.

Problem 3: $\int_0^\infty \frac{\arctan x}{x\sqrt{x}} dx = \int_0^1 \dots + \int_1^\infty \dots$; Is $\int_0^1 \frac{\arctan x}{x\sqrt{x}} dx$ convergent?
 convergent (P2) ≥ 0 on $(0, 1]$, cont.

$L = \lim_{\substack{x \rightarrow 0 \\ x > 0}} (x-0)^p \cdot f(x) = \lim_{\substack{x \rightarrow 0 \\ x > 0}} \frac{x^p \arctan x}{x\sqrt{x}} = \lim_{x \rightarrow 0} \frac{\arctan x}{x} = 1$
 $p = \frac{1}{2} \Rightarrow L = 1$; $p < 1 \Rightarrow \text{convergent}$
 conv. + conv. = conv.

Problem 4: $\int_1^\infty \frac{(\ln x)^4}{(x-1)^3} dx = \int_1^2 \frac{(\ln x)^4}{(x-1)^3} dx + \int_2^\infty \frac{(\ln x)^4}{(x-1)^3} dx$ (2 in 1)

I₁: $L = \lim_{x \rightarrow 1} (x-1)^p \cdot \frac{(\ln x)^4}{(x-1)^3} = \lim_{x \rightarrow 1} \left(\frac{\ln x}{x-1} \right)^4 = 1$
 $\lim_{x \rightarrow 1} \frac{\ln x}{x-1} = 1$ ($\lim_{y \rightarrow 0} \frac{\ln(1+y)}{y} = 1$)
 $x = y+1, x-1 = y$
 $p = -1 \Rightarrow L = 1$
 $\alpha < 0: \lim_{x \rightarrow \infty} x^\alpha \ln x = \lim_{x \rightarrow \infty} \frac{\ln x}{x^{-\alpha}} = \lim_{x \rightarrow \infty} \frac{1/x}{(-\alpha)x^{-\alpha-1}} = -\frac{1}{\alpha} \lim_{x \rightarrow \infty} \frac{1}{x^{-\alpha}} = -\frac{1}{\alpha} \cdot \frac{1}{+\infty} = 0$

$p < 1$ (use C2) $\Rightarrow I_1$ convergent.

I₂: $L = \lim_{x \rightarrow \infty} x^p \cdot \frac{(\ln x)^4}{(x-1)^3} = \lim_{x \rightarrow \infty} \frac{x^p}{(x-1)^3} \cdot \lim_{x \rightarrow \infty} x^{p-3} \cdot (\ln x)^4 = \lim_{x \rightarrow \infty} x^{\frac{p-3}{4}} \cdot \ln x$
 $\lim_{x \rightarrow \infty} x^\alpha \cdot \ln x = \begin{cases} \infty, & \alpha \geq 0 \\ 0, & \alpha < 0 \end{cases} \Rightarrow L = \begin{cases} \infty, & \alpha \geq 0 \Leftrightarrow p \geq 3 \\ 0, & \alpha < 0 \Leftrightarrow p < 3 \end{cases}$
 Use C1; let $p \in (1, 3)$
 $L = 0, p > 1 \Rightarrow \Rightarrow I_2$ is convergent
 conv + conv = conv $\Rightarrow I$ is convergent

Problem 5: $\underbrace{\int_0^{\infty} \frac{e^{-1/x}}{x^3} dx}_I$; $\frac{1}{x} = t$; $x = \frac{1}{t}$; $dx = -\frac{1}{t^2} dt$

$$I = \int_{\infty}^0 e^{-t} \cdot t^3 \cdot \left(-\frac{1}{t^2}\right) dt = \int_0^{\infty} e^{-t} \cdot t \cdot dt = \int_0^{\infty} \frac{t}{e^t} dt$$

$$L = \lim_{t \rightarrow \infty} t^p \cdot \frac{t}{e^t} = \lim_{t \rightarrow \infty} \frac{t^{p+1}}{e^t} = 0, \forall p \in \mathbb{R} \quad \frac{t^3}{e^t} \rightarrow 0 \text{ (use (H) }^3)$$

Use (C1) : $L=0$, let $p > 1$ (ex: $p=2$)

$\Rightarrow$ convergence.

Theorem 4 (C3). Let $f : [a, b) \rightarrow \mathbb{R}_+$ be locally integrable. Let $p \in \mathbb{R}$ such that

$$\lim_{\substack{x \rightarrow b \\ x < b}} (b-x)^p \cdot f(x) = L \in [0, \infty].$$

(i) If $p < 1$ and $L \neq \infty$, then $\int_a^{\infty} f(x) dx$ is convergent.

(ii) If $p \geq 1$ and $L \neq 0$, then $\int_a^{\infty} f(x) dx$ is divergent.